MTH345 Sample Final (100 pts + EC 20 pts)

Please print your name clearly. A penalty may be applied if your name is difficult to read.

Last Name: _____

First Name: _____

Instructions:

- There are nine problems. The points for each problem are indicated next to the question.
- To earn full credit, you must show all necessary work. If it is unclear which assumptions or theorems may be used without proof, write out as much of your reasoning as possible.
- This test is closed-notes, and no computing devices (including calculators) are allowed.

MTH345 Sample Final

P1 (10 pts)
For an event A, let I_A denote the indicator random variable of A, defined by:

$$I_A = \begin{cases} 1, & \text{if } A \text{ occurs,} \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

For any two events A and B with $P(A) > 0$, show that:

$$\operatorname{Cov}[I_A, I_B] = P(A)\big(P(B \mid A) - P(B)\big). \tag{2}$$

P2 (15 pts)
Show that:

$$\log p_\theta(\boldsymbol{x}) \geq \mathrm{E}_{\boldsymbol{Z}\sim q_\phi(\boldsymbol{z}|\boldsymbol{x})}\left[\log p_\theta(\boldsymbol{x} \mid \boldsymbol{Z})\right] - \mathcal{D}_{\mathrm{KL}}\big(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \parallel p(\boldsymbol{z})\big). \tag{3}$$

In this problem, you may use Jensen's inequality without proof: for any convex function ψ,

$$\psi(\mathrm{E}[X]) \leq \mathrm{E}[\psi(X)],$$

where equality holds if and only if X is constant or ψ is linear.

P3 (15 pts)
Let $p(\boldsymbol{x})$ and $q(\boldsymbol{y})$ be two probability density functions on $\mathbb{R}^d$, and let $k(\boldsymbol{x}, \boldsymbol{x}') = \varphi(\boldsymbol{x})^\top \varphi(\boldsymbol{x}')$ be a kernel function with feature map $\varphi : \mathbb{R}^d \to \mathbb{R}^m$. Define the kernel means $\boldsymbol{\mu}_p := \mathrm{E}_{\boldsymbol{X}\sim p}[\varphi(\boldsymbol{X})]$ and $\boldsymbol{\mu}_q := \mathrm{E}_{\boldsymbol{Y}\sim q}[\varphi(\boldsymbol{Y})]$, and define $\mathrm{MMD}_k^2(p, q) := (\boldsymbol{\mu}_p - \boldsymbol{\mu}_q)^\top (\boldsymbol{\mu}_p - \boldsymbol{\mu}_q)$. Show that for mutually independent random vectors $\boldsymbol{X}, \boldsymbol{X}' \overset{\text{i.i.d.}}{\sim} p$ and $\boldsymbol{Y}, \boldsymbol{Y}' \overset{\text{i.i.d.}}{\sim} q$:

$$\begin{aligned}\mathrm{MMD}_k^2(p, q) = \mathrm{E}_{(\boldsymbol{X},\boldsymbol{X}')\sim p(\boldsymbol{x})p(\boldsymbol{x}')}\left[k(\boldsymbol{X}, \boldsymbol{X}')\right] - 2\,\mathrm{E}_{(\boldsymbol{X},\boldsymbol{Y})\sim p(\boldsymbol{x})q(\boldsymbol{y})}\left[k(\boldsymbol{X}, \boldsymbol{Y})\right] \\ + \mathrm{E}_{(\boldsymbol{Y},\boldsymbol{Y}')\sim q(\boldsymbol{y})q(\boldsymbol{y}')}\left[k(\boldsymbol{Y}, \boldsymbol{Y}')\right].\end{aligned} \tag{4}$$

P4 (10 pts)
Let $p(\boldsymbol{c}, \boldsymbol{x}) := p(\boldsymbol{c})p(\boldsymbol{x} \mid \boldsymbol{c})$ denote the joint distribution of condition and data, and let $p_{\theta,\psi}(\boldsymbol{c}, \boldsymbol{x}) := p(\boldsymbol{c})p_{\theta,\psi}(\boldsymbol{x} \mid \boldsymbol{c})$ denote the joint distribution induced by the generator. The conditional minimax objective is given by:

$$V(\theta, \phi, \psi) := \mathrm{E}_{(\boldsymbol{c},\boldsymbol{x})\sim p}\left[\log D_\phi(\boldsymbol{c}, \boldsymbol{x})\right] + \mathrm{E}_{(\boldsymbol{c},\boldsymbol{x})\sim p_{\theta,\psi}}\left[\log\left(1 - D_\phi(\boldsymbol{c}, \boldsymbol{x})\right)\right].$$

Assume the discriminator class $\{D_\phi\}$ is sufficiently expressive such that for any (θ, ψ), there exists $\phi^*(\theta, \psi)$ satisfying:

$$D_{\phi^*(\theta,\psi)}(\boldsymbol{c}, \boldsymbol{x}) = \frac{p(\boldsymbol{c}, \boldsymbol{x})}{p(\boldsymbol{c}, \boldsymbol{x}) + p_{\theta,\psi}(\boldsymbol{c}, \boldsymbol{x})}. \tag{5}$$

Show that $\max_\phi V(\theta, \phi, \psi) = 2\mathcal{D}_{\mathrm{JS}}(p \parallel p_{\theta,\psi}) - \log 4$.

P5 (10 pts)
For a given datum $\boldsymbol{x} \in \mathbb{R}^d$, noise vector $\boldsymbol{\epsilon} \in \mathbb{R}^d$, and noise level $\sigma > 0$, let $\tilde{\boldsymbol{x}} = \boldsymbol{x} + \sigma\boldsymbol{\epsilon}$, so that $p(\tilde{\boldsymbol{x}} \mid \boldsymbol{x})$ is $\mathcal{N}(\tilde{\boldsymbol{x}}; \boldsymbol{x}, \sigma^2 \boldsymbol{I}_d)$. Show that the denoising score matching loss

$$\left\| S_\theta(\boldsymbol{x} + \sigma\boldsymbol{\epsilon}, \sigma) - \nabla_{\tilde{\boldsymbol{x}}} \log p(\tilde{\boldsymbol{x}} \mid \boldsymbol{x}) \right\|_2^2$$

can be expressed as:

$$\left\| S_\theta(\boldsymbol{x} + \sigma\boldsymbol{\epsilon}, \sigma) + \boldsymbol{\epsilon}/\sigma \right\|_2^2. \tag{6}$$

P6 (10 pts)
Consider a Fully Visible Sigmoid Belief Network (FVSBN) for $\boldsymbol{X} \in \{0,1\}^d$:

$$X_j \mid \boldsymbol{X}_{<j} = \boldsymbol{x}_{<j} \sim \text{Ber}\left(\mu_\theta^{(j)}(\boldsymbol{x}_{<j})\right), \quad j = 1, \ldots, d, \tag{7}$$

where $\mu_\theta^{(j)}(\boldsymbol{x}_{<j}) = \text{sig}\left(\alpha^{(j)} + (\boldsymbol{\beta}^{(j)})^\top \boldsymbol{x}_{<j}\right)$ for $j \geq 2$, and $\mu_\theta^{(1)} = \text{sig}(\alpha^{(1)})$, with $\text{sig}(u) := 1/(1+\exp(-u))$.

a. (5 pts) Show that the log-likelihood of a single sample $\boldsymbol{x} \in \{0,1\}^d$ decomposes into a sum of d binary cross-entropy terms:

$$\log p_\theta(\boldsymbol{x}) = \sum_{j=1}^{d} \Big(x_j \log \mu_\theta^{(j)}(\boldsymbol{x}_{<j}) + (1 - x_j) \log\left(1 - \mu_\theta^{(j)}(\boldsymbol{x}_{<j})\right)\Big). \tag{8}$$

b. (5 pts) Derive the gradient of $\log p_\theta(\boldsymbol{x})$ with respect to the weight vector $\boldsymbol{\beta}^{(j)}$ for any $j \in \{2, \ldots, d\}$.

P7 (20 pts)
a. (10 pts) Verify that

$$f(t) = \frac{1}{2}\left(t \log t - (t+1) \log\left(\frac{t+1}{2}\right)\right) \tag{9}$$

satisfies $f(1) = 0$ and is strictly convex on $\mathbb{R}^+$.

b. (10 pts) Let $\mathcal{D}_{\mathrm{JS}}(\cdot \parallel \cdot)$ denote the Jensen-Shannon divergence. Show that

$$\mathcal{D}_{\mathrm{JS}}(f_1 \parallel f_2) = \int_{\mathbb{R}^d} f_2(\boldsymbol{x}) f\left(\frac{f_1(\boldsymbol{x})}{f_2(\boldsymbol{x})}\right) d\boldsymbol{x}. \tag{10}$$

P8 (10 pts)
Let $\mathrm{ELBO}(\theta, \phi; \boldsymbol{x}) := \log p_\theta(\boldsymbol{x}) - \mathcal{D}_{\mathrm{KL}}\big(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \parallel p_\theta(\boldsymbol{z} \mid \boldsymbol{x})\big)$. Show that:

$$\mathrm{ELBO}(\theta, \phi; \boldsymbol{x}) = \mathrm{E}_{\boldsymbol{Z} \sim q_\phi(\boldsymbol{z}|\boldsymbol{x})} \big[\log p_\theta(\boldsymbol{x} \mid \boldsymbol{Z}) \big] - \mathcal{D}_{\mathrm{KL}}\big(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \parallel p(\boldsymbol{z})\big). \tag{11}$$

P9 (EC 20 pts)
Let $p(\boldsymbol{x})$ and $q(\boldsymbol{x}')$ be probability density functions on $\mathbb{R}^d$, and let $\Pi(p,q)$ denote the set of all joint distributions (couplings) $\pi(\boldsymbol{x},\boldsymbol{x}')$ whose marginal densities are p and q. For a metric $d(\cdot,\cdot)$ on $\mathbb{R}^d$, the 1-Wasserstein distance is defined as:

$$W_1(p,q) := \inf_{\pi\in\Pi(p,q)} \int_{\mathbb{R}^d\times\mathbb{R}^d} d(\boldsymbol{x},\boldsymbol{x}')\pi(\boldsymbol{x},\boldsymbol{x}')\,d\boldsymbol{x}\,d\boldsymbol{x}'. \tag{12}$$

a. (10 pts) Assuming strong minimax duality ($\inf_{\pi\geq 0}\sup_{f,g} = \sup_{f,g}\inf_{\pi\geq 0}$), show that:

$$W_1(p,q) = \sup_{f(\boldsymbol{x})+g(\boldsymbol{x}')\leq d(\boldsymbol{x},\boldsymbol{x}')} \left(\int_{\mathbb{R}^d} f(\boldsymbol{x})p(\boldsymbol{x})\,d\boldsymbol{x} + \int_{\mathbb{R}^d} g(\boldsymbol{x}')q(\boldsymbol{x}')\,d\boldsymbol{x}' \right). \tag{13}$$

b. (10 pts) When $d(\boldsymbol{x}, \boldsymbol{x}') = \|\boldsymbol{x} - \boldsymbol{x}'\|$ for some norm $\|\cdot\|$, prove the Kantorovich-Rubinstein duality:

$$\begin{aligned} W_1(p, q) &= \sup_{f(\boldsymbol{x})+g(\boldsymbol{x}')\leq d(\boldsymbol{x},\boldsymbol{x}')} \left(\int_{\mathbb{R}^d} f(\boldsymbol{x}) p(\boldsymbol{x})\, d\boldsymbol{x} + \int_{\mathbb{R}^d} g(\boldsymbol{x}') q(\boldsymbol{x}')\, d\boldsymbol{x}' \right) \\ &= \sup_{\mathrm{Lip}(f)\leq 1} \left(\int_{\mathbb{R}^d} f(\boldsymbol{x}) p(\boldsymbol{x})\, d\boldsymbol{x} - \int_{\mathbb{R}^d} f(\boldsymbol{x}) q(\boldsymbol{x})\, d\boldsymbol{x} \right), \end{aligned} \tag{14}$$

where $\mathrm{Lip}(f) := \sup_{\boldsymbol{x}\neq\boldsymbol{x}'} \frac{|f(\boldsymbol{x})-f(\boldsymbol{x}')|}{\|\boldsymbol{x}-\boldsymbol{x}'\|}$ denotes the Lipschitz constant of f.